INTERACTIVE ROADMAP

Beginner Data Analyst: Final 12-Week Roadmap

From complete beginner to job-ready junior analyst in 6-7 hours a week (72-84 hours). Prepared 14 October 2026 by a Lead Business Analyst, synthesising Roadmaps A, B and C.

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How to use this PDF. Contents lines, week numbers and resource names are clickable links. Section 11 has tickable checkboxes and note fields, so open the file in a PDF reader such as Adobe Acrobat, Preview or a browser to fill them in and save your progress. The bookmark panel also lists every section.

1. Executive summary

This roadmap takes a complete beginner to a job-ready junior data analyst in 12 weeks. It combines the strongest elements of the three source roadmaps:

- **From B:** every resource tagged by cost, honest verification flags, a public GitHub portfolio and a peer-review step.
- **From A:** a dedicated dashboard week, a defined portfolio project brief and a final stakeholder presentation.
- **From C:** deeper SQL (CTEs, CASE WHEN, CSV export) spread over three weeks.

Two additions from a business-analysis perspective: every project starts from a **business question** (not a dataset), and every phase ends with a **plain-language insight**, because employers hire analysts to inform decisions, not just to run queries.

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2. Who this is for

- Complete beginner with basic computer skills
- 6-7 hours per week, ideally in 2-3 sessions
- Prefers free resources (paid options are flagged, never required)
- Target roles: Junior Data Analyst, Business Analyst, Reporting Analyst

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3. Design decisions

Decision	Choice	Why
Sequence	Spreadsheets, SQL, Python, Statistics, Visualization, Dashboard, Portfolio	Matches how most analyst jobs work: data lives in databases, so SQL comes before Python
Statistics timing	Week 8, after Pandas	Learners apply it to data they have already cleaned (A and B), not abstractly (C)
SQL depth	3 weeks	The most commonly tested skill in analyst interviews
Dashboard tool	Tableau Public by default; Power BI optional	Tableau Public is free and cross-platform; Power BI Desktop is free but Windows-only
Portfolio	2 weeks plus a continuous GitHub habit	Evidence of work matters more than certificates

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4. The 12-week plan

Click a week number to jump to its tracker row in Section 11.

Week	Focus	What you do	Completion criterion
Week 1	**Spreadsheets I: cleaning**	Google Sheets or Excel. Sorting, filtering, TRIM, remove duplicates, data types, formatting	Clean a messy 500+ row file and document every change
Week 2	**Spreadsheets II: analysis**	Formulas, XLOOKUP/VLOOKUP, SUMIFS, pivot tables, basic charts	Pivot summary answering 3 business questions
Week 3	**SQL I: querying**	SELECT, WHERE, ORDER BY, LIMIT, GROUP BY, aggregates	SQLBolt lessons 1-12 done and 10 queries written unaided
Week 4	**SQL II: combining data**	INNER/LEFT JOIN, subqueries, HAVING	Answer a 3-table question with JOINs, no copied solution
Week 5	**SQL III: advanced**	CTEs, CASE WHEN, intro to window functions, CSV export	One multi-step CTE query exported for Week 6

Week	Focus	What you do	Completion criterion
Week 6	**Python and Pandas basics**	Variables, lists, dictionaries, loops, functions; load, select, filter, group in Pandas	Week 5 export loaded and a grouped summary produced
Week 7	**Data cleaning and EDA**	Missing values, duplicates, types, outliers, merging, distributions	Cleaned dataset plus notebook with 5 documented findings
Week 8	**Statistics for analysts**	Mean, median, standard deviation, distributions, correlation, sampling, confidence intervals, intro to hypothesis testing	Plain-words written explanation of one statistic and its decision impact
Week 9	**Visualization**	Matplotlib and Seaborn; choosing the right chart; labels, colour, clarity	5 charts, each with a one-line takeaway headline
Week 10	**Dashboarding**	Tableau Public (or Power BI on Windows, or a spreadsheet dashboard): filters, KPIs, layout	Published interactive dashboard, 3+ KPIs and 2 filters
Week 11	**Portfolio build**	Question, data, SQL, Python, dashboard (see Section 6)	Repo with cleaned data, SQL, notebook and dashboard link
Week 12	**Polish, review, present**	README, limitations, peer review, 3-5 minute presentation, CV and LinkedIn update	Peer feedback addressed; presentation given to a real person

Weekly rhythm (about 6.5 hours): 2 hrs learn, 3 hrs practise, 1 hr to write up what you learned and commit it to GitHub. Skipping the write-up is the most common reason beginners cannot explain their own work in interviews.

Slip rule: if you fall behind, do not compress later weeks. Use Week 11 as your buffer and shorten the project scope instead. Weeks 5, 6 and 7 are the most likely to overrun.

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5. Resources and cost

Every item is tagged. Please verify availability and pricing yourself, as free tiers and course access change.

Skill area	Resource (click)	Cost
Spreadsheets	Google Sheets Help	Free
Spreadsheets	GCF Global	Free
Spreadsheets	Microsoft Excel Help	Free docs; Excel needs a licence (Sheets is a free substitute)
SQL	SQLBolt	Free
SQL	Kaggle Learn: Intro to SQL, Advanced SQL	Free
SQL	Mode SQL Tutorial	Free (verify current access)
Python and Pandas	Kaggle Learn: Python, Pandas, Data Cleaning	Free
Python and Pandas	pandas documentation	Free
Statistics	Khan Academy Statistics and Probability	Free
Visualization	Matplotlib docs	Free
Visualization	Seaborn docs	Free
Dashboards	Tableau Public	Free (published work is public)
Dashboards	Power BI, Microsoft Learn	Free Desktop app (Windows only; verify sharing limits)
Broad courses	freeCodeCamp	Free

Skill area	Resource (click)	Cost
Broad courses	Coursera	Audit free (verify); certificate paid and optional
Datasets	Kaggle Datasets	Free
Datasets	data.gov	Free
Portfolio	GitHub	Free

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6. Capstone project brief

Default project: Retail Sales Analysis Dashboard. Learners wanting a different topic may choose a public dataset if they can write the problem statement below first.

Step 1: write the business question. One paragraph covering who the decision-maker is, what decision they face and what success looks like. Example: "Which product categories and regions should the retail manager prioritise next quarter to grow margin?"

Step 2: deliver these artefacts

- Raw data and cleaned data, clearly separated
- SQL file with at least one CTE and one CASE WHEN
- Documented Jupyter notebook (cleaning, EDA, one statistical check)
- Published dashboard with KPIs and filters
- README: question, data source, method, key findings, recommendations
- **Limitations section:** data gaps, assumptions, what you would do with more time
- 3-5 minute presentation aimed at a non-technical stakeholder

Step 3: peer review. Share the GitHub repo with at least one peer or mentor and record the feedback you acted on.

Quality bar: a hiring manager should understand the problem, the answer and the recommendation within 2 minutes of opening your README.

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7. Skills checklist

By Week 12 you should be able to demonstrate:

1. **Spreadsheets:** clean data, build lookups and pivot tables
2. **SQL:** JOINs, subqueries, CTEs, CASE WHEN, aggregation
3. **Python:** write scripts, use Pandas to clean, merge and summarise data
4. **Statistics:** describe distributions, interpret correlation, explain uncertainty
5. **Visualization:** choose appropriate charts and label them clearly
6. **Dashboarding:** build an interactive KPI dashboard
7. **Communication:** turn analysis into a recommendation for a non-technical audience
8. **Professional practice:** use GitHub, write a README, document assumptions and limitations

Tick these off in Section 11.

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8. Job-readiness gates

Use these checkpoints to decide whether to move on or repeat a phase.

After	Gate
After Week 2	You can clean a file and explain every step to someone else

After	Gate
After Week 5	You can answer a business question from a 3-table database using SQL alone
After Week 8	You can take a raw CSV to a cleaned dataset with documented findings
After Week 10	You can publish a dashboard a manager could use without your help
After Week 12	You can present your project in under 5 minutes and handle follow-up questions

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9. After week 12

- **Apply from Week 10 onward.** Do not wait to feel ready.
- **Interview practice:** 2-3 SQL problems per week on platforms such as StrataScratch or LeetCode (verify free-tier limits), plus rehearsing your project story using Situation, Task, Action, Result.
- **Deepen:** window functions, A/B testing, one domain (finance, marketing or operations) and optionally a recognised certification.
- **Keep one project live:** refresh your dashboard with new data monthly to show ongoing practice.

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10. Risks and mitigations

Risk	Mitigation
Tutorial hopping without building	Every week has a deliverable; no deliverable means the week is not complete
Python overwhelm in Weeks 6-7	Stay in Pandas; skip advanced Python such as classes and decorators
Sensitive data on a public dashboard	Use only public datasets
Resource availability changes	Each skill has at least two alternatives in Section 5
Motivation dips around Weeks 5-7	Pair up with a study partner and share weekly commits on GitHub

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11. Interactive progress tracker

Tick the box when you have met the criterion, then type the date finished. Save the file to keep your progress.

Weekly milestones

Done	Week	Completion criterion	Date done
☐	1	**Spreadsheets I: cleaning:** Clean a messy 500+ row file and document every change	
☐	2	**Spreadsheets II: analysis:** Pivot summary answering 3 business questions	
☐	3	**SQL I: querying:** SQLBolt lessons 1-12 done and 10 queries written unaided	
☐	4	**SQL II: combining data:** Answer a 3-table question with JOINs, no copied solution	
☐	5	**SQL III: advanced:** One multi-step CTE query exported for Week 6	
☐	6	**Python and Pandas basics:** Week 5 export loaded and a grouped summary produced	
☐	7	**Data cleaning and EDA:** Cleaned dataset plus notebook with 5 documented findings	
☐	8	**Statistics for analysts:** Plain-words written explanation of one statistic and its decision impact	
☐	9	**Visualization:** 5 charts, each with a one-line takeaway headline	
☐	10	**Dashboarding:** Published interactive dashboard, 3+ KPIs and 2 filters	
☐	11	**Portfolio build:** Repo with cleaned data, SQL, notebook and dashboard link	
☐	12	**Polish, review, present:** Peer feedback addressed; presentation given to a real person	

Job-readiness gates

Passed	Gate
☐	**After Week 2:** You can clean a file and explain every step to someone else
☐	**After Week 5:** You can answer a business question from a 3-table database using SQL alone
☐	**After Week 8:** You can take a raw CSV to a cleaned dataset with documented findings
☐	**After Week 10:** You can publish a dashboard a manager could use without your help
☐	**After Week 12:** You can present your project in under 5 minutes and handle follow-up questions

Skills demonstrated

Can do	Skill
☐	**Spreadsheets:** clean data, build lookups and pivot tables
☐	**SQL:** JOINs, subqueries, CTEs, CASE WHEN, aggregation
☐	**Python:** write scripts, use Pandas to clean, merge and summarise data
☐	**Statistics:** describe distributions, interpret correlation, explain uncertainty
☐	**Visualization:** choose appropriate charts and label them clearly
☐	**Dashboarding:** build an interactive KPI dashboard

Can do	Skill
☐	**Communication:** turn analysis into a recommendation for a non-technical audience
☐	**Professional practice:** use GitHub, write a README, document assumptions and limitations

Capstone deliverables

Done	Deliverable
☐	Business question written
☐	Raw and cleaned data in GitHub
☐	SQL file with CTE and CASE WHEN
☐	Documented notebook
☐	Published dashboard
☐	README with limitations
☐	Peer review recorded
☐	Presentation delivered

My notes

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